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Point-wise discriminative auto-encoder with application on robust radar automatic target recognition

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ABSTRACT

Deep learning methods have a wide range of applications on various recognition problems for their ability to extract hierarchical features. However, most of the deep algorithms learn features from both the target area and the background area fulfilled with noise or clutter. In radar target recognition applications, the performance of applying deep learning methods directly on data with noise or clutter background is restricted because high-level features can be polluted by a significant amount of irrelevant patterns in the background. Therefore, a deep learning based model, point-wise discriminative auto-encoder, is proposed in this paper to extract noise and clutter robust features from the target area. We bind the original auto-encoder (AE) with a target area extraction net, which can learn the target area automatically from the raw data, to reduce the influence of the noise or clutter background. Moreover, we take advantage of the label information by adding a supervised constraint in our algorithm to help the target area extraction net learn the target area precisely and to extract discriminative high-level features. And then, our proposed method is applied to both high-resolution range profile (HRRP) data and synthetic aperture radar (SAR) images in radar automatic target recognition (RATR) problems. Experimental results on the measured HRRP data and SAR images show the advantages of our proposed method in real applications.

1. Introduction

Deep learning [1], which is inspired by the human learning system, is a kind of method constructed by multi-layer structures. Many state-of-the-art results have been achieved by deep learning algorithms in various problems [2] such as classification [3] and detection [4] due to the strong nonlinear fitting and hierarchical feature extraction ability of deep learning algorithms. Deep learning models can be categorized into two main research streams: supervised and unsupervised paradigms. Unsupervised models can learn label free features and are used in various areas without manual tagged data. Supervised models take label information into account and extracts discriminative features to improve the performance. Among unsupervised methods, auto-encoder (AE) [5,6] and Restricted Boltzmann Machine (RBM) [7,8] are two widely used deep learning methods [9,10] because they can be learned and extended easily. Convolutional Neural Network (CNN) [11,12] is the most popular supervised structure for its significant performance in image processing problems.

However, most of the deep learning methods neglect the target area and the background area separation in the feature extraction process. Thus, these methods learn hierarchical features from both target area and the noise or clutter background area. Denoising AE (DAE), which is designed for robustness to small irrelevant changes, introduces random noise in the input [13,14]. But the algorithm does not divide the target area from the noise or clutter background and the high-level features may still be polluted by the noise or clutter in the background. Randomly adding noise to the input data is a commonly used data augmentation method in CNN [11,15]. Similar to DAE, it can improve the robustness to small irrelevant changes, but it is not designed for the robustness to the noise or clutter background. Discriminative conditional variational auto-encoder [16] combines conditional variational AE structure with the average profile to learn features which are robust to aspect sensitivity. But it shares the same subspaces and still cannot separate the target area from the noise background. Thus, the performance of applying these deep methods directly on data with the noise or clutter background is not good because high-level features may be polluted by a large amount of irrelevant patterns in the background. Target-aware recurrent attentional network [17] employ the attention mechanism based on RNN to weight up each timestep in the hidden state so as to discover the target area implicitly. Kihyuk proposed an RBM based method, Point-wise Gated Boltzmann Machine (PGBM) [18], which automatically divides the target area from background through introduction of a gating

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mechanism for estimating where task-relevant patterns occurs. And [19] used PGBM for automated robust feature extraction from contrast-enhanced ultrasound videos to classify benign and malignant cervical lymph nodes. Therefore PGBM can achieve high classification accuracy on images with the noise or clutter background.

Although PGBM can learn features from the target area, it operates in an unsupervised manner. The unsupervised model has the advantage of learning label free features, but without label information the performance of extracting discriminative high-level features is restricted compared to supervised models. Moreover, the target area also has strong connection with the label information. In this study, we proposed the point-wise discriminative AE, which extracts noise or cluster robust features from the target area. We bind the original auto-encoder (AE) with a target area extraction net to reduce the influence of the noise or clutter background, and introduce a supervised constraint into our algorithm, which is designed to learn discriminative features without using pixel-wise label commonly used in segmentation networks such as FCN [43], Deep Mask [44], Mask R-CNN [45].

The main contributions of this paper are summarized as follows:

- We proposed a deep learning based model named point-wise discriminative AE to extract noise and cluster robust features from the target area instead of the whole area with the noise or clutter background.
- Compared to PGBM, we use a specially designed target area extraction net to learn the target area instead of single variable learned from alternative iterations. The target area extraction net and the whole network are learned from gradient descent, which is easier to train and find the optimal solution.
- The supervised constraint helps our network to learn more discriminative features compared to PGBM. Since the target area also has strong connection with label information, the target area extraction net extracts the target area more precisely with the supervised constraint.

Radar data can be obtained in all weather and full-time, and they have significant values in both commercial and military areas. High resolution range profile (HRRP) [20,21] data and synthetic aperture radar (SAR) images are two commonly used radar data for radar automatic target recognition (RATR). An HRRP is the amplitude of the coherent summations of the complex time returns from target scatterers, representing the projection of the echoes from the target scattering centers onto the radar line-of-sight (RLOS). HRRP data contain large amount of structure signatures of the radar target, such as scatterer distribution and target size. Different from the HRRP data, SAR processing can produce high resolution imagery containing 2-D information [24–28]. Besides, SAR images contains both shape information and scattering information, presenting a better representation of the objects. Thus, SAR images are becoming a useful tool for RATR [26–28].

In real applications, the measured HRRP data and SAR images are usually polluted by the noise and clutter. Experimental results in [22,23] have shown that the classification performance of HRRP data dramatically deteriorates under the high noise level. For a certain noise power and a certain radar power, the signal to noise ratio (SNR) is directly related to the distance between the target and radar. Therefore, the classification distance of the radar in real applications can be increased by noise robustness classification methods. For SAR images, clutters also influence both the detection and the classification results significantly [24,25]. Moreover, in real applications with non-cooperative targets, the SNR and the signal to clutter ratio (SCR) of the test samples are usually unknown and different from that of the training samples [29]. In that case, it is very difficult to extract noise and clutter robust features via common algorithms. Therefore, learning noise and clutter robust features is of significant value in real applications. In actual large or complex scene applications such as in urban area, the target detection process is generally required first. Nevertheless, this paper focuses on target recognition from target chips such as MSTAR dataset which consists of many vehicle target chips. We mainly deal with the low SNR/SCR problem in the recognition stage. With applications on the measured HRRP and SAR radar data, our proposed point-wise discriminative AE can extract noise and clutter robust features and achieves good performance.

The rest of the paper is organized as follows. In Section 2 deep learning and AE models are described. In Section 3, our point-wise discriminative AE structure is proposed. Experimental results compared to other methods on measured radar data are shown in Section 4 followed by the conclusions in Section 5.

2. Brief overview of deep learning and AE

2.1. Deep learning

As early as 1989, deep learning was proposed from the study of artificial neural networks which have multiple hidden layers [30]. However, due to the lack of suitable training algorithms for deep neural networks in the 1990s and the early 2000s, people could not use deep models widely until Hinton introduced a deep learning method in 2006 [31]. This training algorithm attracted extensive attention from machine learning related research groups [40,41].

Deep learning algorithms involve a series of models, and these models attempt to extract hierarchical features from the input data using deep networks. Typical deep network architectures include belief networks [1], RBM [7,8], AE [5,6], CNN [11,12], Variational Auto-Encoder (VAE) [36], Generative Adversarial Networks (GAN) [37], etc. Among unsupervised deep structures, AE and RBM networks which are commonly used to extract hierarchical features, are layer-wise learned via unsupervised training strategies and can be followed by a supervised fine-tuning step. Among supervised deep structures, CNN is the most widely used model in image processing due to its ability to extract highly nonlinear features. In deep learning algorithms high-level features are learned from low-level ones, and hierarchical features learned by deep structures are suitable for classification. Besides, deep models extract more abstract and complex features [38,39] than traditional shallow models and approximate to highly nonlinear functions better.

2.2. Structure of AE

AE is an unsupervised structure which is commonly used [12]. AE was firstly proposed in the 1980′s [40,41] and then used in various classification problems [10] due to that the AE structure can be learned and extended easily [10]. An ordinary AE structure is commonly composed of an encoder and a decoder as shown in Fig. 1.

During the training stage, the input x is first maps the input to hidden layer and produces the hidden representation $h \in R^{d_h}$ via the encoder of the AE structure. The boxed part of Fig. 1 shows corresponding step of the network, which has the form

$$h = f(x) = s_f(Wx + b_h) \tag{1}$$

where s_f is a nonlinear activation function, normally a logistic function as $sigmoid(z) = \frac{1}{1+e^{-z}}$. Other nonlinear activation functions, such as ReLU [32], can also be used in the nonlinear mapping. The encoder is parameterized by a $d_h \times d_x$ weight matrix W, and a bias vector $b_h \in R^{d_h}$.

Subsequently, a decoder is employed to reconstruct the original input by mapping the hidden units h to an output following the

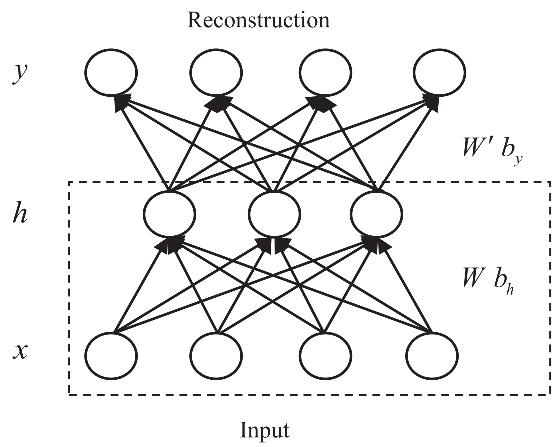

Fig. 1. Structure of a single layer AE, in which the hidden feature '*h*' is learned from input '*x*' and then is reconstructed to '*y*'.

encoder. The form of the reconstruction is as follows:

$$y = g(h) = s_g(W'h + b_y), \tag{2}$$

where s_g is the activation function, which commonly has the form of a linear function or a sigmoid function. The bias vector $b_y \in R^{d_x}$ and the weight matrix W' are the parameters of the decoder. Tied weights are commonly used in AE algorithms, in which $W' = W^T$ where T denotes the transpose operation.

The encoder outputs the hidden layer features and then the decoder is adopted to reconstruct the initial input x from the hidden layer features $f(x)$ by minimizing the following cost function:

$$J_{AE}(\theta) = \sum_{x \in D_n} L(x, y) = \sum_{x \in D_n} L(x, g(f(x))), \tag{3}$$

where L is the reconstruction error, typically the squared error function $L(x, y) = \|x - y\|_2^2$. The parameters of the AE are $\theta = \{W, b_h, b_y\}$ and D_n is the total training set. The gradient descent algorithms can be used to update the parameters.

3. Point-wise discriminative AE

Deep learning is widely used in various areas because it can extract hierarchical features. However most of the deep learning methods neglect considering the target and the background separately in feature learning. When we handle the data with noise or clutter background in real applications, it is necessary for a deep learning method to distinguish semantically different patterns. For example, if a target recognition algorithm can separate the target area from the noise or clutter background area, it may improve its recognition performance. To model this, we propose the point-wise discriminative AE with a specially designed target area extraction net and supervised constraint. Fig. 2 shows the brief structure of the proposed method.

In Fig. 2, the target area extraction part learns the target area from the raw input. The training process includes the pre-train stage and the main network training stage. During the pre-train stage, the target area extraction net is firstly pre-trained with original input data without pixel-wise label. Then the pre-trained weights are initialized and is updated with the whole network. In detail, we use the original input data as the expected output of the target area extraction net during the pre-train stage because pixel-wise label is not at hand in real applications. After the pre-train stage, the chosen target area is sent to the main network to learn discriminative features via the supervised constraint using label information. The three parts of the network is described in detail in the following subsections. After the discriminative features are

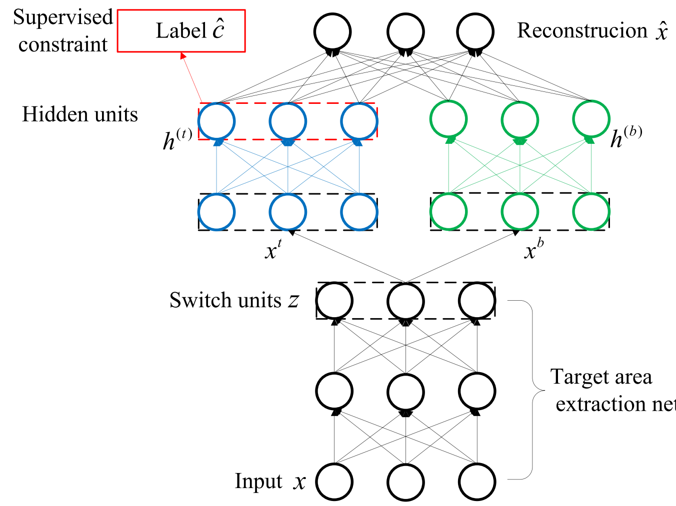

Fig. 2. Structure of our proposed method, Point-wise Discriminative AE. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

learned by our proposed method, we can use classifiers such as support vector machine (SVM) [26] to recognize the target.

3.1. Target area extraction net

The target area extraction net is designed to learn the target area from the raw input. Let X be the training data and x_i $(i = 1, \ldots, N)$ represents the ith training sample, where $x_i \in R^n$ is a column vector and N is the total number of the training data. In Fig. 2, the target area extraction net outputs the switch units $z_i \in R^n$ from each input x_i via a multi-layer feedforward network (here we use two layers as an example):

$$z_i = f_{znet}(x_i) \tag{4}$$

where f_{znet} represents the nonlinear mapping of the network which uses the sigmoid function as nonlinear activation functions in both two layers. More specifically, the Eq. (4) can be represented in Eq. (5):

$$\begin{aligned} g_i &= f_1(W_1 x_i + b_1) \\ z_i &= f_2(W_2 x_i + b_2) \end{aligned} \tag{5}$$

where f_1 and f_2 are nonlinear activation function, g_i denotes the hidden representation of input x_i, $\theta = \{W_1, b_1, W_2, b_2\}$ are the parameters of the 2-layer feedforward network. Each element of the output $z_{i,k} \in (0, 1)$ $(k = 1, \ldots, n)$ is the probability of whether the corresponding $x_{i,k}$ is target or background. That is, the larger $z_{i,k}$ is, the more likely $x_{i,k}$ is supposed to be the part of the target area instead of the background area. Therefore, when $z_{i,k} > 0.5$ the probability of the corresponding $x_{i,k}$ regarding as the target is greater than that as the background, when $z_{i,k} < 0.5$ the probability of the corresponding $x_{i,k}$regarding as the target is less than that as the background. We can regard the value 0.5 as the threshold which decide about being part of background or target. After the target area extraction net outputs the switch units, we obtain the target area x_i^t by

$$x_i^t = z_i \otimes x_i \tag{6}$$

where the operator $\otimes$ denotes element-wise multiplication, i.e. $z_i \otimes x_i = [z_{i,1}x_{i,1}, \ldots, z_{i,n}x_{i,n}]^T$. Also, we can obtain the background area x_i^b by

$$x_i^b = (1 - z_i) \otimes x_i \tag{7}$$

In real applications, the target area is usually much smaller than the background area. That means the switch units z_i is sparse. Thus, we add the sparse constraint to the switch units z_i in the loss function:

$$J_{sparse} = \sum_{i=1}^{N} \rho \log \frac{\rho}{\sum_{k=1}^{n} z_{i,k}} + (1-\rho) \frac{1-\rho}{1-\sum_{k=1}^{n} z_{i,k}} \quad (8)$$

where ρ is a constant representing the degree of sparsity.

3.2. Supervised constraint

The supervised constraint with red lines in Fig. 2 helps the whole network to learn discriminative features and helps the target area to learn the target area precisely. Only the target area is mapped to the semantic label because the semantic label information is strongly related with the target area not the background. In the part with blue lines Fig. 2, after obtaining the target area x_i^t by the target area extraction net, x_i^t is encoded to the hidden units $h_i^t \in R^{m_t}$:

$$h_i^t = sigmoid(W_t x_i^t + b_t). \quad (9)$$

where $W_t \in R^{n \times m_t}$ is the weight matrix, and $b_t \in R^{m_t}$ is the bias vector. The hidden units h_i^t is supposed to be the discriminative features which is directly associated to the label of the input x_i. Therefore, the hidden units h_i^t is used to predict the label of x_i via another nonlinear mapping:

$$\hat{c}_i = sigmoid(W_l h_i^t + b_l). \quad (10)$$

In (10), $\hat{c}_i \in R^l$ is the prediction label; $W_l \in R^{n \times l}$ is the weight matrix; and $b_l \in R^l$ is the bias vector. The supervised constraint ensures that the prediction label $\hat{c}_i$ is closed to the exact label c_i of the input x_i via the squared error loss function:

$$J_{label} = \sum_{i=1}^{N} \left\| c_i - \hat{c}_i \right\|^2. \quad (11)$$

3.3. Reconstruction

The reconstruction of AE is an unsupervised feature extraction method which can provide good generalization performance [34]. The reconstruction part in our algorithm reconstruct the raw input x_i from the target area x_i^t and the background area x_i^b together. As shown in the part with green lines in Fig. 2, the background area x_i^b is encoded to the hidden units $h_i^b \in R^{m_b}$ by:

$$h_i^b = sigmoid(W_b x_i^b + b_b). \quad (12)$$

where $W_b \in R^{n \times m_b}$ is the weight matrix, and $b_b \in R^{m_b}$ is the bias vector. Then the hidden units h_i^t and h_i^b are used together to reconstruct the input $x_i$by a nonlinear decoder:

$$y_i = sigmoid\left(W_r \begin{bmatrix} h_i^t \\ h_i^b \end{bmatrix} + b_r \right). \quad (13)$$

In (13), y_i is reconstruction from the hidden units; $W_r \in R^{(m_t+m_b)\times n}$ is the weight matrix; and $b_r \in R^n$ is the bias vector. The reconstruction error is defined as the squared error in the loss function:

$$J_{reconstruction} = \sum_{i=1}^{N} \|x_i - y_i\|^2. \quad (14)$$

The total loss function of our proposed method is combined of all the three parts mentioned above:

$$J_{PGAE} = \lambda_1 J_{label} + \lambda_2 J_{sparse} + J_{reconstruction}. \quad (15)$$

The parameters of the whole structure are learned by gradient descent algorithms, similar to the original AE.

Table 1
Parameters of three airplanes and radar in the experiment.

Radar parameters	Center frequency: 5520 MHz Bandwidth: 400MHz		
Planes	Length (m)	Width (m)	Height (m)
Yark-42	36.38	34.88	9.83
Cessna Citation S/II	14.40	15.90	4.57
An-26	23.80	29.20	9.83

3.4. Data augmentation

Data augmentation is commonly used in CNN [11,15] to improve the performance. With the augmented data, CNN has more training samples which can partially solve overfitting and provide good generalization performance. Commonly used data augmentation methods in CNN include: shift, mirror, rotation, randomly injecting noise, etc.

In this paper, the main problem we focus on is to extract noise and clutter robust features from the measured radar data with the background fulfilled with noise or clutter. For non-cooperative targets, the SNR and the SCR of the test samples are usually unknown and different from that of train samples [29]. Therefore, the data augmentation method used in our algorithm is different from those commonly used in CNN. Here we augment the training samples by adjusting their SNRs or SCRs according to the real physical principles of simulating the radar data. For the HRRP data, each training sample is added by different complex Gaussian noises to obtain training samples with different SNRs. For example, in Fig. 3, an original HRRP sample, which is about 30 dB SNR, is augmented to 5 dB, 10 dB, 15 dB and 20 dB SNR. All the training samples are used to train the network together.

For the SAR images, we use image quilting [35] to create different clutter samples and then inject them to the background of the original images. Thus, training samples with different SCRs are obtained. Fig. 4 shows an example of SAR image augmentation, in which 5 dB, 10 dB, 15 dB and 20 dB SCR images are created from an original SAR image for training the network.

4. Experiments

4.1. Data description

In our study, both the HRRP data and SAR images are tested using our proposed Point-wise discriminative AE.

4.1.1. Measured HRRP data

The measured HRRP data used in our study contains two different types of yet planes, Cessna Citation S/II and Yark-42, and a propeller plane An-26. Table 1 shows the detailed parameters of the three airplanes and the parameters of the radar. The HRRP data are segmented, where each segment contains 25,600 samples. In the experiments of the HRRP data, the elevation angles of the test data should be different from those of the training data to test the generalization performance of our proposed method, and the training data should cover almost all the target-aspect angles to train the model. Therefore, we chose the second and fifth segments of Yark-42, the sixth and seventh segments of Cessna Citation S/II, the fifth and the sixth of An-26 as training data, and other data segments are chosen as test data in the experiments. It is mentioned that the division of training data and test data is the same with [16].

4.1.2. Measured 3-target SAR images

The 3-target MSATR dataset is a measured SAR dataset which contains two types of tanks, T72 and BMP2, and an armored vehicle, BTR70. Fig. 5 shows the SAR and the optical image examples of the three targets. The aspect angles of each target are covered

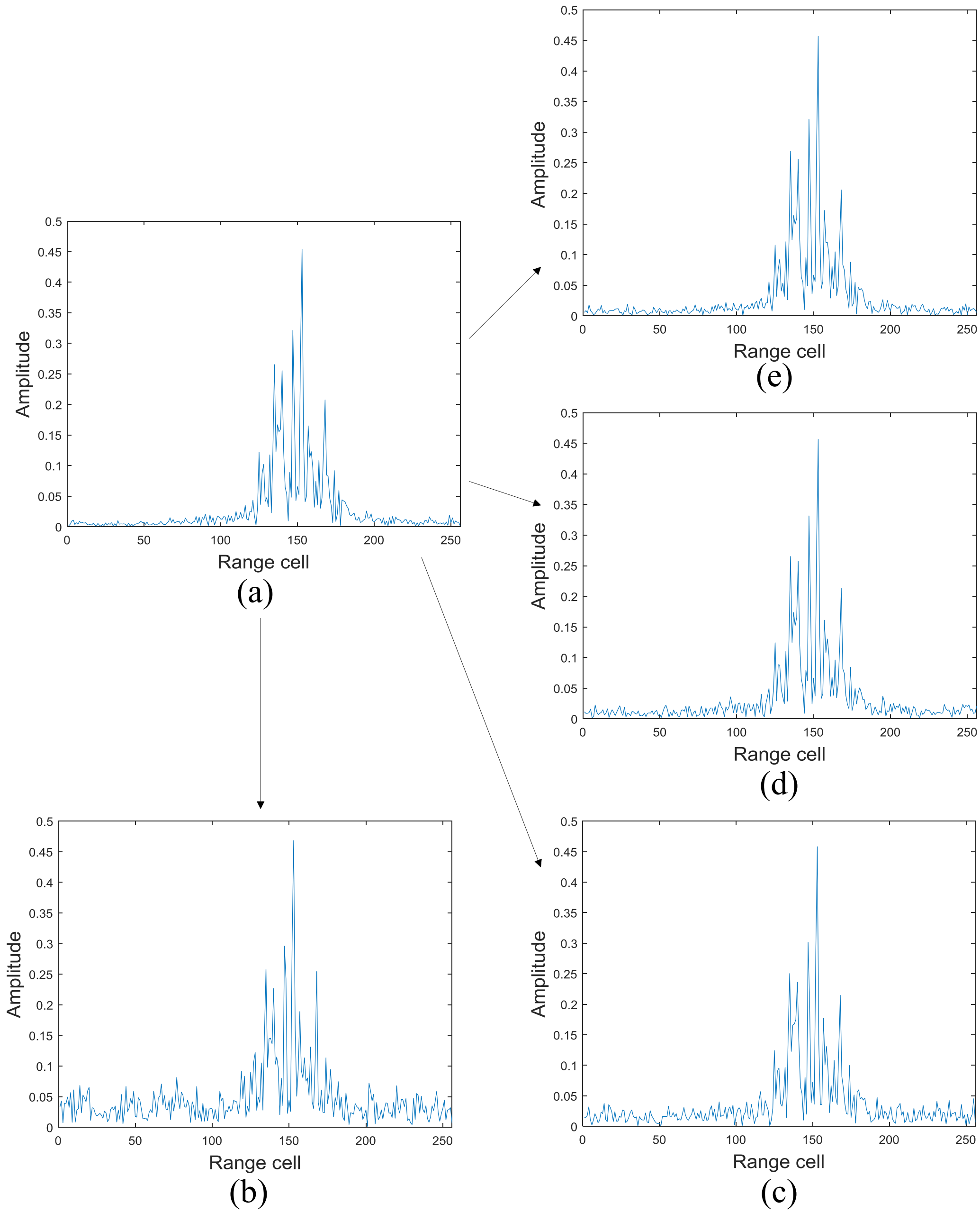

Fig. 3. An example of HRRP data augmentation. (a) is the original HRRP training sample. (b)–(e) are the augmented training samples with 5 dB, 10 dB, 15 dB and 20 dB SNR respectively.

Table 2
Type and number of training and test samples for 3-target data.

Dataset	BMP2			BTR70	T72		
	C21	9566	9563	C71	132	S7	812
Training samples (17°)	233	0	0	233	232	0	0
Test samples (15°)	196	196	195	196	196	191	195

from 0° to 360°, and each target has two different views at depression angle 15° and depression angle 17° In the experiments of the SAR images, the data at depression 17° are used for training and the data at depression 15° are used for test.

Table 2 lists the detailed number of the training and test SAR images of the three targets. As shown in Table 2, BMP2 and T2 both have three different serial numbers. In our experiments on the measured SAR images, only the data of serial number BMP2-9563, BTR70-C71 and T72-132 at depression angle 17° are used for training; while the data of all serial numbers at depression angle 15° are used for test.

4.1.3. Measured 10-target SAR images

Besides the 3-target data, the 10-target dataset contains the data from another 7 ground vehicles. The optical imagery examples of these ten types of targets are shown in Fig. 6. Similar to

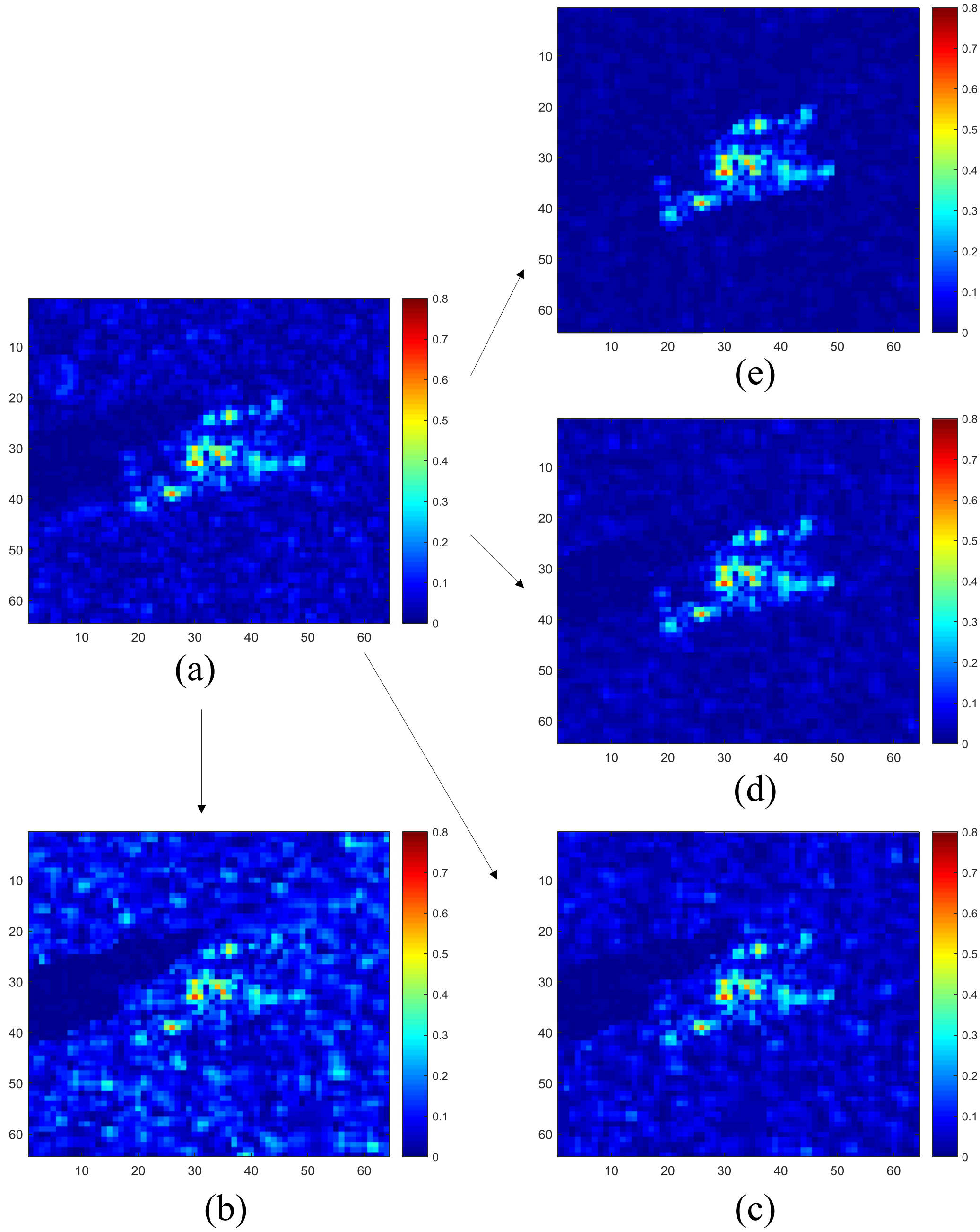

Fig. 4. An example of SAR data augmentation. (a) is the original SAR training image. (b)–(e) are the created training images with 5 dB, 10 dB, 15 dB and 20 dB SCR respectively.

the recognition experiments on 3-target data, we use the images at depression angle 17° without variants as the training data, and all images at depression angle 15° as the test data in the 10-target experiments. Some details of the training and test samples of the 10-target data are listed in Table 3.

4.2. Experiments on HRRP data

4.2.1. Experiment setup

In HRRP data experiments, a 2-layer target area extraction net with 128 hidden units in is used. The main network has 625 discriminative hidden units (h^t) and other 625 hidden units mapped from the background (h^b). The supervised constraint weight λ_1 is set to 3 and the sparsity weight of the target area extraction net λ_2 is set to 0.0001. After extracting discriminative features, a linear SVM is used for classification. Training and test samples are both augmented using the augmentation method mentioned in Section 3.4. HRRP data with 5 dB, 10 dB, 15 dB, 20 dB and 30 dB SNR are used for training our proposed method. HRRP data with 0 dB, 5 dB, 7.5 dB, 10 dB, 12.5 dB, 15 dB, 17.5 dB, 20 dB, 25 dB and 30 dB SNR are used for test.

4.2.2. Recognition results

In this subsection, the recognition results on the HRRP data with different SNRs is shown. We compare the performance of our proposed method with traditional HRRP recognition methods,

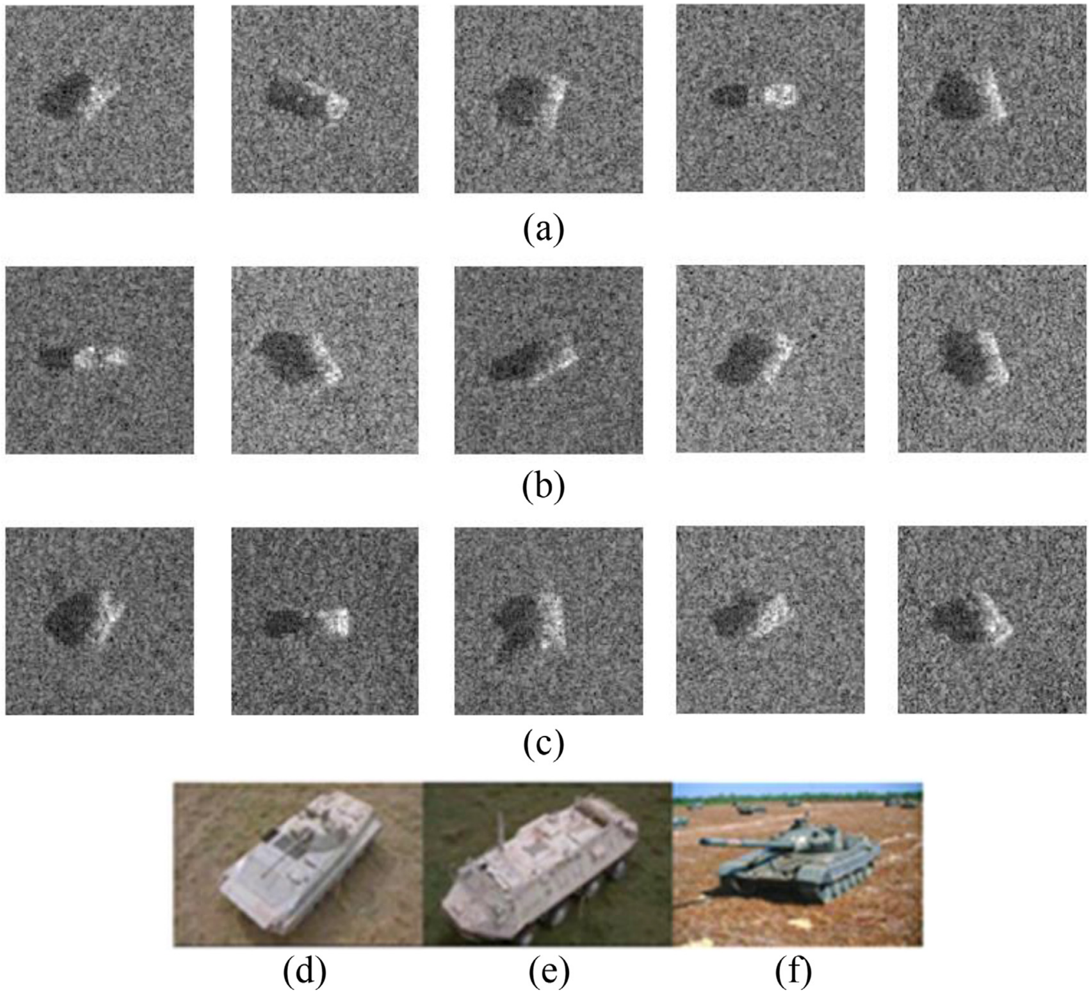

Fig. 5. SAR and optical images of the 3-target data to be identified. (a) gives some SAR imagery examples of BMP2 and (d) is its optical image. (b) gives some imagery examples of BTR70 and (e) is its optical image. (c) gives some imagery examples of T72 and (f) is its optical image.

Table 3
Type and number of training and test samples for 10-target data.

Dataset	BMP2	BTR70	T72	BTR60	2S1	BRDM2	D7	T62	ZIL131	ZSU234
Training samples (17°)	233 (C21)	233	232 (132)	255	299	298	299	299	299	299
Test samples (15°)	587 (C21,9566,9563)	196	582 (132,S7,812)	195	274	274	274	273	274	274

directly applying linear SVM to the HRRP data (Original HRRP data) and principle component analysis (PCA) with linear SVM. Results of commonly used deep learning methods, original AE followed by linear SVM (original AE), RBM followed by linear SVM (RBM), DAE followed by linear SVM, 1-D CNN [33] and FDCVAE [16], are also compared. In addition, we compared the recognition results of our proposed method with PGBM, which is also followed by linear SVM as classifier (PGBM). The results of applying our data augmentation method to PGBM is also compared to show the effectiveness of our proposed point-wise discriminative AE. Accuracies of different methods are shown in Table 4.

As shown in Table 4, our proposed method yields the highest accuracies in low SNR values among all the compared methods because of the target area extraction net and the supervised constraint. Especially, the training samples of 7.5 dB, 12.5 dB, 17.5 dB and 25 dB SNRs are not included in the augmented training data. Accuracies of these test data do not drop obviously compared to those of other test data (5 dB, 10 dB, 15 dB, 20 dB and 30 dB) in our proposed method. That means our proposed method can deal with the condition that the SNR of the test data is different to those of the training samples. Compared to traditional HRRP recognition methods (Original HRRP data and PCA), our proposed method performs better on all test HRRP data with different SNRs. Our proposed method also yields higher accuracies than those of common deep learning methods, original AE, RBM, and DAE. Although DAE introduces random noise in the input, which is designed for robustness to small irrelevant changes, its high-level features are still polluted by the noise or clutter background. 1-D CNN yields the highest accuracies on HRRP test data with high SNRs (25 dB and 30 dB) for its great ability to learn non-linear hierarchical features. But the performance of CNN on low SNR test data drops significantly because the data augmentation method of randomly adding noise cannot reduce the influence of the noise or clutter background. Compare to PGBM, our proposed method has higher accuracies on all test HRRP data with different SNRs. That shows the effectiveness of our proposed method to deal with

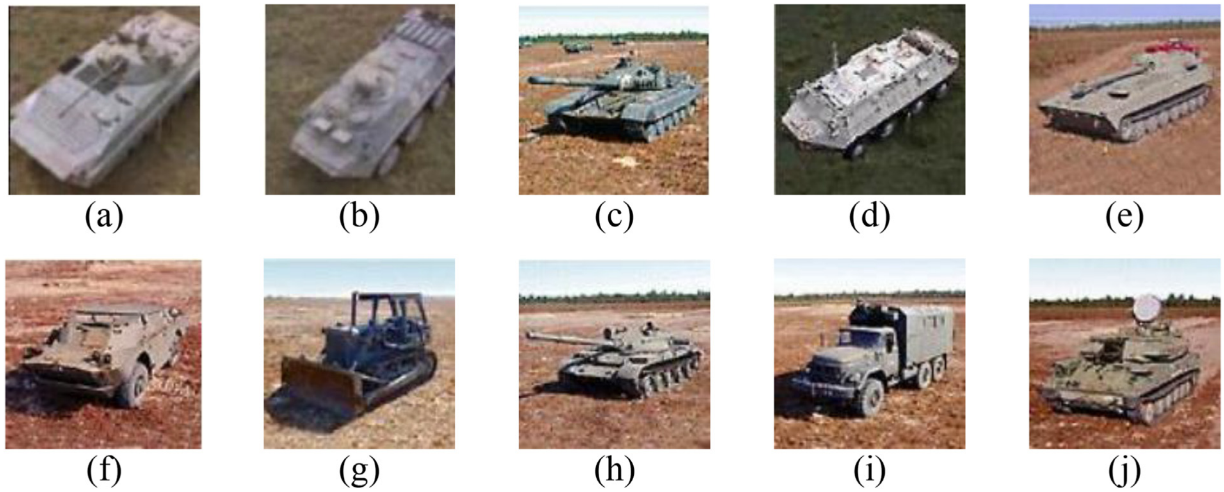

Fig. 6. Optical images of the 10-target MSTAR data.(a)–(j) are imagery examples of BMP2, BTR70, T72, BTR60, 2S1, BRDM2, D7, T62, ZIL131, ZSU234 respectively.

Table 4
Accuracies of different methods on HRRP data. Accuracies in bold type are the highest accuracy of each column.

SNR										
Methods	0dB	5dB	7.5dB	10dB	12.5dB	15dB	17.5dB	20dB	25dB	30dB
Proposed method	**62.04%**	**71.12%**	**73.19%**	74.54%	**76.81%**	78.33%	**80.85%**	82.83%	85.06%	85.46%
Original HRRP data	39.85%	44.52%	47.90%	51.84%	55.12%	57.60%	59.94%	69.06%	80.83%	85.77%
PCA	25.13%	43.31%	47.19%	51.50%	54.83%	57.75%	59.25%	66.04%	83.12%	84.52%
Original AE	42.67%	48.79%	50.46%	51.85%	54.33%	56.46%	57.83%	63.29%	80.17%	81.73%
RBM	42.23%	48.08%	52.58%	56.29%	58.50%	60.19%	62.04%	71.23%	81.44%	81.69%
DAE	43.15%	46.52%	49.81%	51.77%	52.77%	52.33%	52.57%	57.33%	68.42%	72.48%
1D-CNN	51.59%	59.50%	62.00%	63.48%	63.80%	65.83%	70.86%	79.96%	**91.51%**	90.63%
PGBM	40.12%	47.60%	53.62%	58.00%	59.56%	60.19%	64.29%	72.79%	81.87%	81.78%
PGBM (data augmentation)	59.58%	65.04%	69.04%	71.21%	73.27%	74.67%	75.87%	76.38%	77.34%	78.27%
FDCVAE[16]	61.61%	69.58%		**75.39%**		**81.04%**		**87.66%**	90.66%	**92.85%**

the noise or clutter background. Moreover, recognition accuracies of PGBM adopted our data augmentation method are still lower than our proposed point-wise discriminative AE. It demonstrates that the target area extraction net and the supervised constraint in our proposed method contribute to learn the noise and clutter robust features. Factorized discriminative conditional variational auto-encoder (FDCVAE) [16] combines conditional variational AE structure with the average HRRP profile to learn features which are robust to aspect sensitivity. Our proposed method achieves higher accuracies than FDCVAE [16] in the low SNR case (0 dB and 5 dB), which demonstrates better noise robustness of our method compared to FDCVAE [16].

4.2.3. Effect of the target area extraction net

In this subsection, we show the effect of the target area extraction net and compare the switch units z learned by the target area extraction net of our proposed method with that of PGBM [16]. For 1D HRRP data, the length of the support area corresponds to the target size. In our study, the learned switch units can reflect the length of the support area and the target size can be further estimated based on the learned switch units. The shape of the switch units is automatically learned via our proposed method. In Fig. 7, the switch units learned from a low SNR (5 dB), a medium SNR (12.5 dB) and a high SNR (30 dB) HRRP test sample are compared. The three HRRP test samples are augmented from one same test sample, and we can see the switch units learned from the three test samples are similar by our proposed method. That means the target area extraction net can work with test samples of different SNR. In addition, as shown in Fig. 7, the switch units learned by our proposed method cover the target area better than those learned by PGBM especially in the case of low test.

SNR (5 dB). The switch units learned by our proposed method is approximately saturate, which is mainly due to the sparse constraint in Eq. (8). Specially, the switch units of the background area are almost 0 instead of around 0.5 compared to PGBM. That demonstrates the target area extraction net in our proposed method learns the target area more precisely and the precise target area helps the algorithm to extract discriminative deep features.

4.3. Experiments on 3-target SAR images

4.3.1. Experiment setup

In 3-target MSTAR experiments, a multi-layer target area extraction net is used. Two layer target area extraction net with 1985 hidden units, three layer target area extraction net with 1000–400 hidden units and four layer target area extraction net with 1200–1000–800 hidden units are used. We use the best result under different layers as our final result. The main network has 625 discriminative hidden units (h^t) and other 625 hidden units mapped from the background (h^b). The supervised constraint weight λ_1 is set to 1 and the weight of the target extraction net sparsity λ_2 is set to 0.0002. After extracting discriminative features, a linear SVM is employed for classification. Training and test samples are both augmented using the augmentation method mentioned in Section 3.4. SAR images with 5 dB, 10 dB, 15 dB and 20 dB SCR are used for training our proposed method. SAR images with 0 dB, 5 dB, 7.5 dB, 10 dB, 12.5 dB, 15 dB, 17.5 dB and 20 dB SCR are used for test.

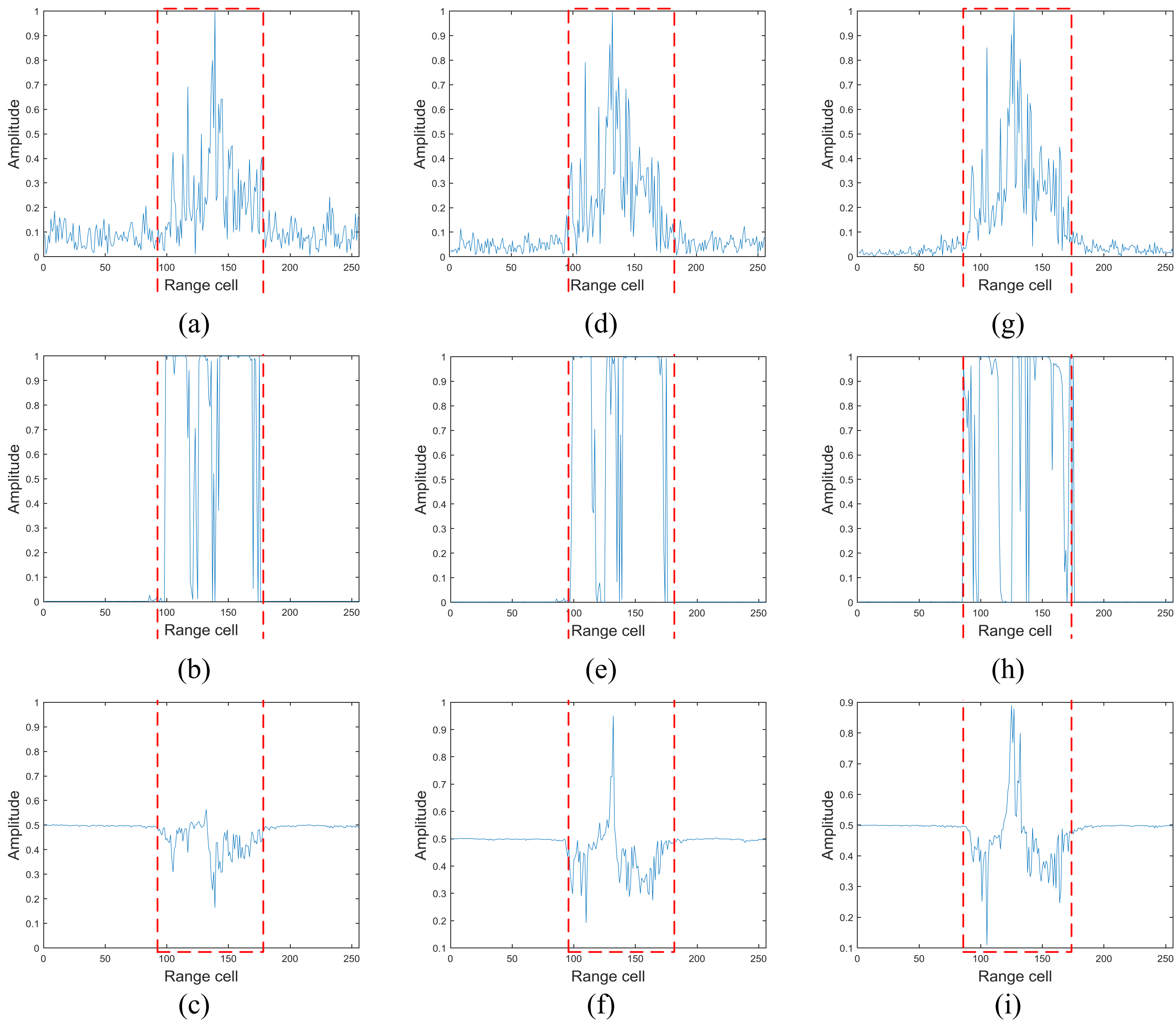

Fig. 7. The switch units z learned from test HRRP samples with different SNR by our proposed method and PGBM. (a) is the 5 dB SNR HRRP test sample. (b) and (c) are the switch units z learned by our proposed method and PGBM respectively. (d) is the 12.5 dB SNR HRRP test sample. (e) and (f) are the switch units z learned by our proposed method and PGBM respectively. (g) is the 30 dB SNR HRRP test sample. (h) and (i) are the switch units z learned by our proposed method and PGBM respectively. Areas within red lines are the target area. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 5
Accuracies of different methods on 3-target MSTAR data. Accuracies in bold type are the highest accuracy of each column.

SCR									
Methods	0dB	5dB	7.5dB	10dB	12.5dB	15dB	17.5dB	20dB	Avg.
Proposed method	**70.21%**	**74.21%**	**77.58%**	74.54%	**82.27%**	80.00%	82.71%	82.93%	**78.06%**
Original SAR images	11.21%	14.36%	19.49%	31.94%	36.70%	47.91%	65.20%	82.20%	38.63%
PCA	12.10%	14.36%	15.53%	24.47%	38.83%	48.79%	64.98%	82.12%	37.65%
Original AE	33.18%	35.90%	42.78%	48.64%	57.64%	74.80%	84.10%	82.27%	57.41%
RBM	36.42%	42.78%	45.27%	52.79%	64.62%	78.75%	84.62%	85.20%	61.31%
DAE	38.21%	43.22%	50.18%	52.97%	55.31%	63.30%	84.10%	87.99%	59.41%
VGG net [12]	35.28%	38.81%	34.78%	34.34%	51.35%	76.82%	**91.77%**	**94.59%**	57.22%
A-ConvNets[42]	30.33%	37.00%	44.03%	53.55%	64.54%	77.14%	88.21%	90.40%	60.65%
PGBM	31.66%	37.36%	48.79%	58.53%	70.70%	**82.56%**	85.42%	86.15%	62.65%
PGBM (data augmentation)	62.19%	69.60%	74.51%	**77.14%**	79.19%	79.27%	80.89%	81.69%	75.56%

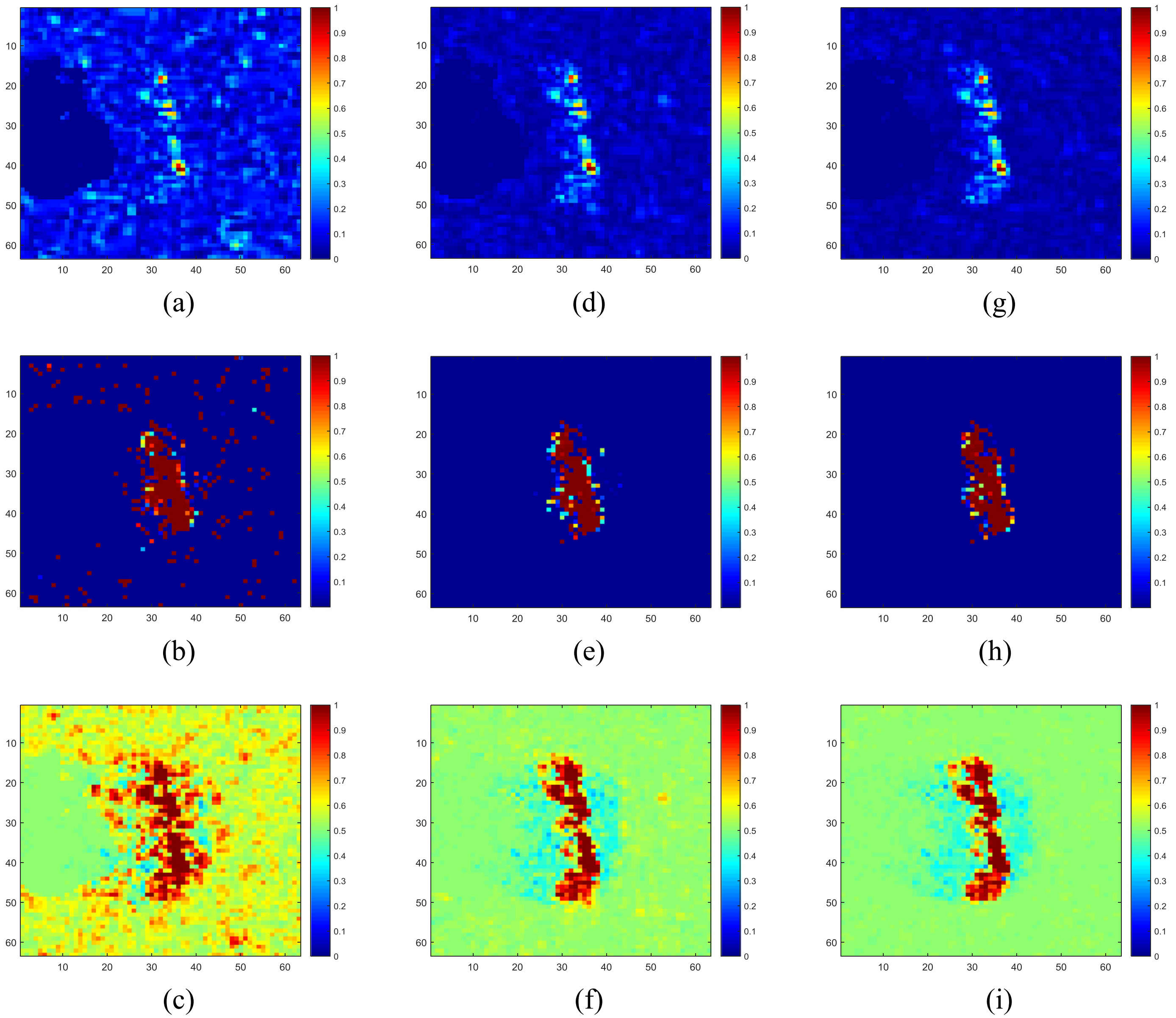

Fig. 8. The switch units z learned from MSTAR test images with different SCR by our proposed method and PGBM. (a) is the 5 dB SCR MSTAR test image. (b) and (c) are the switch units z learned by our proposed method and PGBM respectively. (d) is the 12.5 dB SCR MSTAR test image. (e) and (f) are the switch units z learned by our proposed method and PGBM respectively. (g) is the 30 dB SCR MSTAR test image. (h) and (i) are the switch units z learned by our proposed method and PGBM respectively. .

Table 6
Accuracies of different methods on 10-target MSTAR data. Accuracies in bold type are the highest accuracy of each column.

SCR								
Methods	5dB	7.5dB	10dB	12.5dB	15dB	17.5dB	20dB	Avg.
Proposed method	**63.75%**	**67.25%**	**69.62%**	**70.65%**	**71.28%**	71.53%	72.21%	**69.47%**
Original SAR images	9.15%	11.43%	19.51%	29.19%	48.49%	56.32%	55.27%	32.77%
PCA	10.43%	12.93%	20.76%	29.91%	46.89%	55.95%	55.73%	33.23%
Original AE	11.27%	12.74%	17.61%	27.19%	40.34%	49.27%	48.89%	29.62%
RBM	13.89%	17.42%	23.42%	32.41%	48.05%	57.54%	57.63%	35.77%
DAE	10.27%	11.90%	16.11%	28.32%	48.83%	53.29%	51.33%	31.44%
VGG net [12]	23.13%	28.38%	31.50%	39.31%	53.44%	78.13%	**90.10%**	49.14%
PGBM	14.20%	22.01%	35.52%	52.76%	68.43%	76.70%	77.61%	49.60%
PGBM (data augmentation)	48.64%	54.23%	58.47%	60.47%	59.72%	57.7584%	55.66%	56.42%
A-ConvNets[42]	17.52%	28.61%	30.75%	41.92%	51.53%	**80.25%**	87.89%	48.35%

4.3.2. Recognition results

In this subsection, we show the recognition results on the 3-target MSTAR data with different SCRs. We compare the performance of our proposed method with traditional recognition methods, directly applying linear SVM to the MSTAR data (Original SAR image) and PCA followed by linear SVM (PCA). Results of commonly used deep learning methods, original AE followed by linear SVM (original AE), RBM followed by linear SVM (RBM), DAE followed by linear SVM and CNN (VGG net [12], A-ConvNets [42]), are also compared. In addition, we compared the recognition results of our proposed method with PGBM, which is also followed by linear SVM as classifier (PGBM). The results of applying our data augmentation method to PGBM is also compared to show the effectiveness of our proposed point-wise discriminative AE. Accuracies of different methods on the 3-target MSTAR data are shown in Table 5.

As shown in Table 5, our proposed method yields the highest average accuracy among all the compared methods because of the target area extraction net and the supervised constraint. Especially, the training samples of 0 dB, 7.5 dB, 12.5 dB and 17.5 dB SCRs are not included in the augmented training data. Accuracies of these test data do not drop obviously compared to those of other test data (0 dB, 5 dB, 10 dB, 15 dB and 20 dB). That means our proposed method can also deal with the condition that the SCR of the test SAR images is different to those of the training images. Compared to traditional recognition methods (Original HRRP data and PCA), our proposed method performs better on all test data with different SCRs. Our proposed method also yields higher average accuracy than that of common deep learning methods, original AE, RBM and DAE. Especially our proposed method has great advantage on test images with low SCRs (0dB-15 dB) compared to original AE, RBM and DAE. According to the recognition results, introducing random noise in the input of DAE cannot work with the noise or clutter background of SAR images. CNN yields the highest accuracies on SAR images with high SCRs (17.5 dB and 20 dB) for its great ability to learn hierarchical features. But the performance of CNN on low SCR test images drops obviously because the data augmentation method of randomly adding noise cannot reduce the influence of the noise and clutter background on SAR images. Compared to PGBM and PGBM with our data augmentation method, our proposed point-wise discriminative AE yields higher average accuracy. That shows the effectiveness of our proposed method to learn the noise and clutter robust features because of the target area extraction net and the supervised constraint in our proposed method.

4.3.3. Effect of the target area extraction net

In this subsection, we show the effect of the target area extraction net and compare the switch units z learned by the target area extraction net of our proposed method with that of PGBM. In Fig. 8, the switch units learned from a low SCR (5 dB), a medium SCR (12.5 dB) and a high SCR (20 dB) MSTAR test image are compared.

The three MSTAR test image are augmented from one same image, and we can see the switch units learned from the three test samples are similar by our proposed method. That means the target area extraction net can work with different SCR test images. In addition, as shown in Fig. 8, the switch units learned by our proposed method are concentrated on the target area and cover the target area better in the case of low SCR (5 dB) compared to those learned by PGBM. Specially, the switch units of the background area are almost 0 instead of around 0.5. That demonstrates the target area extraction net in our proposed method learns the target area more precisely and the target area helps the algorithm to extract discriminative deep features.

4.4. Experiments on 10-target SAR images

4.4.1. Experiment setup

Analogous to the 3-target MSTAR experiments, a 2-layer target area extraction net with 1985 hidden units is used. The main network has 625 discriminative hidden units (h^t) and other 625 hidden units mapped from the background (h^b). The supervised constraint weight λ_1 is set to 1 and the weight of the target extraction net sparsity λ_2 is set to 0.0004. After extracting discriminative features, a linear SVM is employed for classification. Training and test samples are both augmented using the augmentation method mentioned in Section 3.4. SAR images with 5 dB, 10 dB, 15 dB and 20 dB SCR are used for training our proposed method. SAR images with 5 dB, 7.5 dB, 10 dB, 12.5 dB, 15 dB, 17.5 dB and 20 dB SCR are used for test.

4.4.2. Recognition results

In this subsection, we show the recognition results on the 10-target MSTAR data with different SCRs. The comparison is the same with those listed in Table 5. Accuracies of different methods on the 10-target MSTAR data are shown in Table 6.

As shown in Table 6, compared to traditional recognition methods (Original HRRP data and PCA), our proposed method performs better on all test data with different SCRs. Our proposed method also yields higher average accuracy than that of common deep learning methods, original AE, RBM and DAE. Especially our proposed method has great advantage on test images with low SCRs (5dB-15 dB) compared to original AE, RBM and DAE. According to the recognition results, introducing random noise in the input of DAE cannot work with the noise or clutter background of SAR images. CNN (VGG net [12] and A-ConvNets [42]) yields the highest accuracies on SAR images with high SCRs (17.5 dB and 20 dB respectively) for its great ability to learn hierarchical features. But the performance of CNN on low SCR test images drops obviously because the data augmentation method of randomly adding noise cannot reduce the influence of the noise and clutter background on SAR images. Compared to PGBM and PGBM with our data augmentation method, our proposed point-wise discriminative AE yields higher average accuracy. That shows the effectiveness of our proposed method to learn the noise and clutter robust features because of the target area extraction net and the supervised constraint in our proposed method.

5. Conclusion

In this paper, we proposed a noise and clutter robust deep learning algorithm — point-wise discriminative AE — for real recognition applications of radar data with noise or clutter background. Specifically, our proposed algorithm combined AE structure with the specially designed target area extraction net and the supervised constraint to automatically divide the target area from the background and further extract discriminative features. The experimental results show the effect of our point-wise discriminative AE on both HRRP data and SAR images compared with other methods.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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